

MANIKANDAN GUNASEELAN

☎ +1(720)270-7993 | ✉ Manikandan.Gunaseelan@colorado.edu | in [manikandan-gunaseelan](https://www.linkedin.com/in/manikandan-gunaseelan) | 🎧 [manig1729](https://open.spotify.com/playlist/manig1729)

EDUCATION

University of Colorado, Boulder

MS in Computer Science

August 2025 – May 2027

GPA : 3.9/4.0

BITS Pilani, K K Birla Goa Campus

BE in Electronics and Communication Engineering

August 2018 – July 2022

GPA : 8.05/10

TECHNICAL SKILLS

Languages: Python, Java, C/C++, JavaScript, SQL, Bash, HTML/CSS

Libraries & Frameworks: PyTorch, TensorFlow, Spark Streaming, Flask, Matplotlib, pytest

Cloud & Big Data: Google Cloud Platform (GCP), Apache Spark, Hadoop, Google Kubernetes Engine (GKE)

Messaging & Storage: Redis, MinIO, PostgreSQL, SQLite

DevOps & Tools: Docker, Kubernetes, Git, Postman, Linux

Networking: Cisco IOS-XR, gRPC, NETCONF, gNMI, Cisco Telemetry

AI Services: AWS Bedrock, AWS Transcribe

EXPERIENCE

Cisco Systems

January 2022 – June 2025

Software Engineer II

- Architected and deployed **Combinatorial Interaction Testing** for networking protocols, reducing test run times by **77%** and increasing pre-release bug detection by **31%**
- Resolved a critical **gNMI Telemetry** synchronization issue for a high-value client (**Google**), securing business continuity for a **\$10M+** account
- Developed **Python-based telemetry APIs** for multi-collector operations, enabling high-throughput data ingestion and integrated **Matplotlib** visualizations for real-time network health monitoring

Software Engineer I

- Authored and maintained **pygnoi**, a Pythonic interface for **gNOI services** adopted by internal network automation teams to simplify **RPC integration** across global frameworks
- Led technical training sessions on **Object-Oriented Design** and **Time Complexity** for a team of 40+, establishing standardized Python coding practices and improving overall codebase maintainability

Technical Undergraduate Intern

- Developed a full-stack dashboard using **Flask** and **JavaScript** to centralize tracking and operational metrics for **60+** automation packages

PROJECTS

TempoMind: Garmin-Spotify Heart-Rate Orchestrator | CU Boulder | GitHub

Fall 2025

- Engineered a **real-time ingestion pipeline** using **Garmin BLE** and **Redis Streams** to synchronize heart-rate telemetry with **Spotify Web Playback** for dynamic BPM-to-song matching
- Architected a **multi-container environment** via **Docker Compose** (Redis, MinIO, SQLite) using **Socket.IO** to maintain sub-second state synchronization across ingestion, playback, and feedback UIs
- Implemented a robust storage layer using **Redis** for transient data streams and **MinIO/SQLite** for persistent telemetry mirroring and session-history deduplication

SemEval-2026: Counterfactual Reasoning for Polarization Detection | SemEval | GitHub

Fall 2025

- Developed transformer-based models (DeBERTa-v3, RoBERTa) to classify English polarized discourse across binary and manifestation-level categories using social media data
- Implemented **Counterfactual Data Augmentation (CDA)** to evaluate model robustness, specifically investigating if counterfactual reasoning improves the detection of nuanced online polarization
- Architected a **Hugging Face** and **PyTorch** pipeline to handle label imbalance and contextual noise, focusing on the mitigation of linguistic biases in automated moderation systems

HACKATHONS

AWS GameDay Hackathon | Amazon @ CU Boulder | Top 5 Finish

September 2025

- Prototyped a **GenAI healthcare application** using **Amazon Bedrock** for RAG-based search and **AWS Transcribe** for automated speech analytics of patient-provider interactions

CERTIFICATIONS

Google IT Support Professional Certificate | Coursera - Google

Introduction to Data Analytics for Business | Coursera - CU Boulder